27 **D** The equation is $I = I_0 \sin(\frac{2\pi t}{T}) = I_0 \sin(\frac{2\pi t}{0.0025}) = I_0 \sin(800\pi t)$

28 **B** $\langle P \rangle = I_{rms}{}^2 R = (I_0/\sqrt{2})^2 R = (2.0/\sqrt{2})^2(200) = 400$ W